

Research Interests

- AI coding agents, agentic AI systems, LLM evaluation and code reasoning, LLM fine-tuning and post-training.
- AI and software engineering, synthetic data generation, software testing (flaky test detection & repair, mutation testing), and genetic algorithms.

My work has been published at multiple top SE venues (FSE, ICSE, ISSTA, TSE, etc.).

Education

- 2022–Present **University of Illinois Urbana-Champaign (USA).**
Ph.D. Candidate in Computer Science (Anticipated graduation: 2027 May)
Co-advisors: [Reyhaneh Jabbarvand](#) and [Darko Marinov](#)
- 2018–2022 **Huazhong University of Science and Technology (China).**
B.Sc. in Computer Science

Publications and Preprints

- [1] Under Review Unlocking Model Potentials Through Adaptive Multi-Agent Scaffolding for Efficient Issue Resolution [\[PDF\]](#) .
2026 **Yang Chen**, Changshu Liu, Aliya Ahmad, Yiheng Zhou, and Reyhaneh Jabbarvand.
- [2] Under Review SENTINEL: Learning Agent Workflow Guardrails from Offline Trajectories.
2026 **Yang Chen**, Jinyang Li, Malinda Dilhara, Vidyashankar Sivakumar, and Mukul Prasad.
- [3] Under Review Narrowing the Complexity Gap in the Evaluation of Large Language Models [\[PDF\]](#) .
2026 **Yang Chen**, Shuyang Liu, and Reyhaneh Jabbarvand.
- [4] Under Review Evaluating Code Reasoning Abilities of Large Language Models Under Real-World Settings [\[PDF\]](#) .
2025 Changshu Liu, Alireza Ghazanfari, **Yang Chen**, and Reyhaneh Jabbarvand.
- [5] FSE Can Old Tests do New Tricks for Resolving SWE Issues? [\[PDF\]](#) .
2026 **Yang Chen**, Toufique Ahmed, Reyhaneh Jabbarvand, and Martin Hirzel.
Proceedings of the ACM International Conference on the Foundations of Software Engineering. Montreal, Canada. July 2026.
- [6] OOPSLA Process-Centric Analysis of Agentic Software Systems [\[PDF\]](#) .
2026 Shuyang Liu, **Yang Chen**, Rahul Krishna, Saurabh Sinha, Jatin Ganhotra, and Reyhaneh Jabbarvand.
OOPSLA 2026. Oakland, California, USA. October 2026.
- [7] ICSE Assessing Coherency and Consistency of Code Execution Reasoning by Large Language Models [\[PDF\]](#) .
2026 Changshu Liu, **Yang Chen**, and Reyhaneh Jabbarvand.
Proceedings of the 48th International Conference on Software Engineering. Rio de Janeiro. Brazil. April 2026.
- [8] TSE CodeMind: Evaluating Large Language Models for Code Reasoning [\[PDF\]](#) .
2026 Changshu Liu, **Yang Chen**, and Reyhaneh Jabbarvand.
IEEE Transactions on Software Engineering. 2026.
- [9] ISSTA Neurosymbolic Repair of Test Flakiness [\[PDF\]](#) .
2024 **Yang Chen** and Reyhaneh Jabbarvand. **[139 patches accepted in real-world projects]**
Proceedings of the 33rd ACM SIGSOFT International Symposium on Software Testing and Analysis. Vienna, Austria. September 2024.

- [10] ICSE-SRC Flakiness Repair in the Era of Large Language Models [\[PDF\]](#) .
 2024 **Yang Chen**. **[2nd Place in ACM Student Research Competition]**
 Proceedings of the 46th International Conference on Software Engineering, Lisbon, Portugal.
 April 2024.
- [11] ISSTA Transforming Test Suites into Croissants [\[PDF\]](#) .
 2023 **Yang Chen**, Alperen Yildiz, Darko Marinov, and Reyhaneh Jabbarvand.
 Proceedings of the 32nd ACM SIGSOFT International Symposium on Software Testing
 and Analysis, Seattle, USA. July 2023.
- [12] SCAM Automated Bug Generation in the Era of Large Language Models [\[PDF\]](#) .
 2025 Ali Reza Ibrahimzada, **Yang Chen**, Ryan Rong, and Reyhaneh Jabbarvand.
 Proceedings of the 25th IEEE International Conference on Source Code Analysis and
 Manipulation. Auckland, New Zealand. 2025.
- [13] ICSE-FTW A Preliminary Study of Fixed Flaky Tests in Rust Projects on GitHub [\[PDF\]](#) .
 2025 Tom Schroeder, Minh Phan, and **Yang Chen**.
 Flaky Test Workshop in Proceedings of the 47th International Conference on Software
 Engineering. Ottawa, Canada. April 2025.
- [14] ICSE-FTW Can ChatGPT Repair Non-Order-Dependent Flaky Tests? [\[PDF\]](#) .
 2024 **Yang Chen** and Reyhaneh Jabbarvand.
 Flaky Test Workshop in Proceedings of the 46th International Conference on Software
 Engineering. Lisbon, Portugal. April 2024.
- [15] ICSE-Demo iPFlakies: A Framework for Detecting and Fixing Python Order-Dependent Flaky Tests
 2022 [\[PDF\]](#) .
 Ruixin Wang, **Yang Chen**, and Wing Lam.
 Demonstration Track, Proceedings of the 44th International Conference on Software
 Engineering, Pittsburgh, USA. May 2022.

Experience

- 2026 **Applied Scientist Intern**, *Amazon Web Services (AWS), CA*.
 Working on LLM coding agent guardrails [2].
- 2025 **Research Scientist Intern**, *IBM Research, NY*.
 Working on LLM coding agents for issue resolution [5].

Selected Honors and Grants

- 2024 Ranked 2nd in the 46th ACM Student Research Competition at ICSE 2024.
- 2024 SIGSOFT CAPS Grants for ICSE 2024 and ISSTA 2024.
- 2023 SIGSOFT CAPS Grants for ISSTA 2023.
- 2022 Outstanding Graduate of Class 2022.

Academic Service

Reviewer: MSR 2024, TSE 2025; Junior PC: MSR 2024; Artifact Evaluation PC: ISSTA
 2024 & 2025; Student Volunteer: ICSE 2022, ISSTA 2023.